

Track 2 — Drug Repurposing for MVA1 (BUB1B compound heterozygosity)

Team: ciberpty · Rare Disease, Real Kid: MVA Hackathon 2026
Repository: <https://github.com/cpu-16/mva-hackathon-2026> · **Methods description:** `methods_track2.md`

Executive summary

We applied five filters to candidate drugs rather than the usual two, and the third one changes the answer.

The candidate the literature points to is **metformin**. It descends from the AICAR hit in the aneuploidy-selective screen of Tang et al. (PMID 21315436), it is approved, and its mechanism is plausible. It fails on pharmacokinetics: plasma concentrations at therapeutic doses are **micromolar**, while complex I inhibition requires **millimolar** (PMID 37343530) — an order-of-magnitude gap of roughly **1000-fold**. We expect metformin to be widely proposed in this track. **It cannot act through complex I inhibition at therapeutic doses**, which is the mechanism the whole proposal rests on.

The candidate that survives is **bortezomib**. Aneuploid cells carry a stoichiometric protein imbalance and compensate by increasing protein degradation, which makes them **preferentially dependent on the proteasome**. Ippolito et al. (2024, *Cancer Discovery*, PMID 39247952) established this across multiple diploid-versus-aneuploid models, recapitulated it in hundreds of cancer cell lines and primary tumours, and found that **aneuploidy level was significantly associated with the response of multiple myeloma patients to proteasome inhibitors** — a human clinical association rather than a pure cell-culture inference.

We state the size of the remaining leap rather than hide it. That evidence is drawn from **cancer** aneuploidy, and myeloma is a plasma-cell neoplasm carrying its own immunoglobulin-driven proteotoxic

load. Whether the same dependency holds in **constitutional, variegated** aneuploidy is an extrapolation, and it is exactly what the experiment in §6 is designed to break.

Bortezomib clears the pharmacokinetic filter with margin: EC50 in highly aneuploid lines is below **40 nM**, while the approved 1.3 mg/m² IV dose reaches a Cmax of **89–120 ng/mL = 231–312 nM** — a ratio of **5.8–7.8**. We quote the whole label interval rather than its ceiling; read it as an order-of-magnitude margin, not a measured number.

It then fails our fourth filter, and we say so plainly: **not for this child, not now**. Bortezomib causes peripheral neuropathy in 18% of paediatric patients (8% motor), and this child already has skeletal muscle atrophy. Outside an active malignancy that risk-benefit does not hold.

What remains is a specific and actionable proposition. Bortezomib does not repair the lesion — nothing restores BubR1 function — but it is the agent that exploits the best-characterised *consequence* of that lesion, and it is the one worth having ready for an MVA patient **with an active tumour**, where the comparator is not “no drug” but conventional cytotoxic chemotherapy. This child is an embryonal rhabdomyosarcoma survivor with a substantial risk of a further embryonal tumour. Our contribution is to have that answer ready — with its evidence, its concentration margin and its stopping rule — before the decision has to be made under pressure.

We also flag an antagonism a combination-minded team could walk into: **mTOR inhibitors would be expected to protect aneuploid cells from proteasome inhibition, not synergise** (PMID 31530568).

1. The variants and what they break

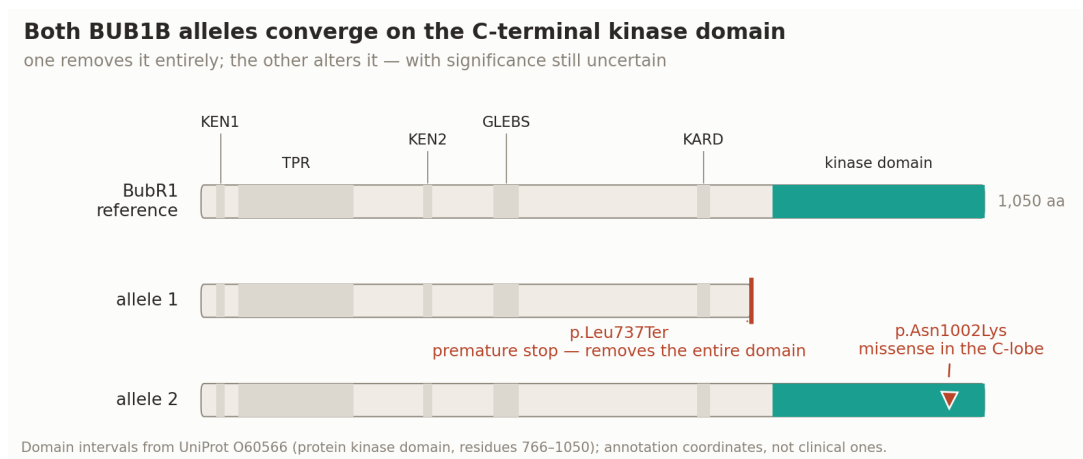
The proband is compound heterozygous in **BUB1B** (NM_001211.6, transcript ENST00000287598.11, MANE):

Variant	Consequence	Exomiser ACMG	ClinVar
c.2210T>G p. (Leu737Ter)	stop_gained	PATHOGENIC (PVS1, PM2_Supporting, PP4_Moderate, PP5_Strong)	533901 — Pathogenic/Likely pathogenic
c.3006T>G p. (Asn1002Lys)	missense	UNCERTAIN_SIGNIFICANCE (PM2_Supporting, PP4_Moderate, BP1)	not present

Population frequency depends on the source, and we report both rather than select the lower one. Exomiser's gnomAD snapshot gives 9.98×10^{-5} (exomes, NFE) for the nonsense and 9.0×10^{-7} for the missense; Ensembl VEP returned 3×10^{-5} and no record respectively. Both are rare; neither is absent.

A note of caution on the missense. ClinVar holds c.3006T>A, a *different* nucleotide substitution producing the same p.Asn1002Lys, classified VUS. That classification does not transfer to our allele: PS1 applies when the known same-amino-acid variant is pathogenic, and here it is uncertain. Our variant is simply not in ClinVar.

BubR1 is a core component of the mitotic checkpoint complex and of kinetochore–microtubule attachment control.



Domain map of BubR1 showing both alleles converging on the C-terminal kinase domain

p.Leu737Ter truncates in the linker N-terminal to the C-terminal kinase fold — UniProt O60566 annotates the protein kinase domain at residues **766–1050** — removing it entirely, about 313 of the protein's 1050 residues. (Whether BubR1 is catalytically active in humans is contested; we call it the kinase domain because that is the current UniProt annotation, and our argument rests on the truncation, not on

catalysis.) It lands 16 residues upstream of **BUBR1^{X753}**, a characterised truncating MVA allele (PMID 31738183) — very nearly the same truncation, not merely a related model.

p.Asn1002Lys falls inside that same domain, in its C-lobe, outside KEN, TPR and GLEBS, and introduces a positive charge. **No published functional assay of this substitution exists.**

The closest characterised missense sits ten residues away, and the mouse work on it carries a warning for us. Sieben et al. modelled the human MVA missense **BUBR1^{L1012P}** in mice and paired it with **BUBR1^{X753}** — the same truncating-plus-missense architecture as our proband, with the missense in the same domain. Those **BubR1^{X753/L1002P}** animals **died prematurely**; the viable genotype paired the missense with a *hypomorphic* allele that still yields some wild-type protein (PMID 31738183).

Our proband is alive carrying a truncating allele. If that result transfers, it constrains p.Asn1002Lys: **it would have to retain more residual function than L1012P does** — a hypomorph rather than a functional null. That is a falsifiable prediction about our own VUS, and the allele-fate arm of the experiment in §6 tests it directly.

Whether the truncated transcript escapes nonsense-mediated decay is unmeasured, and should not be asserted either way without patient RNA.

At the cellular level, cell lines derived from MVA patients with biallelic mutations show “an impaired mitotic checkpoint, chromosome alignment defects, and low overall BUBR1 abundance” (PMID 20516114). BubR1 binds the B56 family of PP2A regulatory subunits through a conserved motif phosphorylated by Cdk1 and Plk1, and counteracts Aurora B at improperly attached kinetochores (Kruse et al., *J Cell Sci* 2013, PMID 23345399); the same B56 partnership opposes Aurora B to give stable kinetochore–microtubule attachment (Xu et al., *Biol Open* 2013, PMID 23789096). The region is commonly called KARD, a name neither source uses.

2. Five filters, not two

Repurposing exercises usually apply two filters: **is it approved?** and **is the mechanism plausible?** Both are necessary; neither is sufficient. Three more decide the outcome.

#	Filter	Question	What it eliminates here
1	Regulatory	Approved for marketing?	AICAR, 17-AAG, reversine, apcin, proTAME, DCZ0415
2	Mechanistic	Does it address the actual lesion?	Agents acting on uninvolved pathways
3	Pharmacokinetic	Is $C_{max} \geq$ the effective in vitro concentration?	Metformin (~1000-fold short)
4	Patient-specific safety	Compatible with <i>this</i> child's comorbidities?	Bortezomib, at present
5	Direction of effect	Does it avoid increasing mis-segregation in survivors?	Anything selecting for unstable clones

Filters 3 and 5 are rarely applied and eliminate the most candidates. Filter 3 is a possibility check, not a demonstration of efficacy: it ignores protein binding, exposure duration and tissue penetration, and a transient post-bolus C_{max} can overstate useful exposure.

3. Applying the filters

3.1 Metformin fails on pharmacokinetics

The chain of reasoning is genuinely attractive and we followed it to the end. Tang et al. screened isogenic trisomic mouse fibroblasts and found selective sensitivity to **AICAR**. AICAR is not approved, so it fails filter 1. Metformin is the marketed stand-in that also produces AMPK activation through complex I inhibition, and it is labelled in paediatrics from age 10. It passes filters 1 and 2.

It fails filter 3 decisively. Sakellakis (2023, PMID 37343530):

“the plasma concentration of metformin remains within the micromolar range when administered at therapeutic doses. While millimolar concentrations are necessary to inhibit complex I activity in isolated mitochondria, there is no evidence supporting the idea that metformin accumulates within the mitochondria.”

Fontaine (2018, PMID 30619086) reviews why that inhibition remains unsettled — that review exists to “gather arguments for and against each hypothesis.” In paediatrics the gap widens: dose-normalised plasma exposure varies **3.3-fold** between children, and “**renal clearance was higher in children compared to the reported data in healthy adults**”, with eGFR significantly affecting oral clearance (PMID 42399684). Metformin is renally cleared, so this child’s nephrocalcinosis makes renal function the parameter that would have to be established first — the source links clearance to eGFR, not to nephrocalcinosis specifically, and we do not stretch it further.

We do not reject metformin out of caution. We reject it with a number — and we state what kind of number it is. For bortezomib, filter 3 is a true $C_{max} / EC50$ ratio against an $EC50$ measured in an aneuploidy-stratified model. For metformin no such $EC50$ exists: no published study gives an aneuploidy-selective effective concentration. AICAR’s aneuploidy selectivity has in fact been reproduced in human cells — a trisomy-7 colonic line is fourfold more sensitive to AICAR than its isogenic diploid counterpart (PMID 22890317). That paper does not test metformin, and no published study gives metformin an aneuploidy-selective effective concentration at all. The 1000-fold figure is therefore a comparison between *concentration ranges* — micromolar plasma against millimolar complex I inhibition — not the same quantity as the bortezomib ratio. It is a coarser instrument, and it still points the same way.

There is a deeper objection, and stating it accurately matters. Tang et al. did **not** restrict themselves to stable trisomies: Figure 6 of that paper tests AICAR and 17-AAG in *Bub1b*^{H/H} hypomorphic MEFs and in *Cdc20*^{AAA} cells — genuine ongoing-instability models with a weak checkpoint. So the screen does reach beyond a fixed extra chromosome.

What it does not reach is **metformin**, which is not tested in those models, and human biallelic BUB1B cells, which are not tested at all. The gap is therefore narrower and more specific than “stable trisomies only”: the marketed stand-in was never evaluated in the instability models, and a mouse *Bub1b*^{H/H} hypomorph is not equivalent to human p.Leu737Ter/p.Asn1002Lys.

3.2 Bortezomib passes filters 1–3, on direct evidence

Mechanism. Aneuploidy imposes a stoichiometric imbalance of protein complexes. Ippolito et al. (2024, PMID 39247952) dissected multiple diploid-versus-aneuploid models and found that aneuploid cells mitigate proteotoxic stress by reducing translation and **increasing protein degradation**, rendering them more sensitive to proteasome inhibition. The finding was recapitulated across hundreds of human cancer cell lines and primary tumours.

The clinically important sentence in that paper is the last one: **aneuploidy level was significantly associated with the response of multiple myeloma patients to proteasome inhibitors**. That is a human signal linking degree of aneuploidy to response to this drug class, which is more than most repurposing arguments have.

We state the size of that clinical signal rather than let a reviewer find it. It rests on small cohorts: with bortezomib as a single agent, 8 patients with a complete response had significantly higher aneuploidy scores than 50 with progressive disease ($p = 0.014$, one-tailed Mann-Whitney); in a bortezomib-plus-chemotherapy-and-dexamethasone cohort, 13 complete responders versus 14 minimal responders ($p = 0.038$); a third cohort trended the same way but had only 2 non-responders. That is a genuine human association built on small numbers — enough to justify running an experiment, not enough to predict an individual patient’s response.

It is also not evidence about MVA1. Myeloma is a plasma-cell neoplasm whose immunoglobulin output imposes a proteotoxic load of its own, and its aneuploidy is clonal, not variegated. MVA1 is a constitutional aneuploidy state — a *different* setting in which the same dependency is plausible and untested. We treat the myeloma association as the reason to run the experiment, not as its result.

The numbers.

	Value	Source
EC50 in highly aneuploid lines (72 h)	< 40 nM — conservative upper bound read from Fig. 6o (5 near-euploid vs 5 highly aneuploid cancer lines, $p = 0.0317$, Mann-Whitney); the figure legend reports no numeric EC50 values, so this is read off the plot	PMID 39247952
C _{max} , 1.3 mg/m ² IV, repeated dosing	89–120 ng/mL = 231–312 nM (label interval; we quote both ends)	FDA label, NDA 021602 s040, §12.3
C _{max} , subcutaneous route	20.4 ng/mL = 53.1 nM	same
Filter 3 ratio (IV)	5.8 – 7.8	—

We use the upper bound of the EC50 range, and we do not treat the 2.4 nM apoptosis figure from the same paper as an EC50.

The subcutaneous route does not pass, and we withdraw the claim that it does. Its nominal ratio is 1.33 on *total* drug, and bortezomib is extensively protein-bound, so free concentration is a fraction of total C_{max}. A margin of 1.33 does not survive that correction. Only the IV route clears filter 3 — and even there, the ratio is computed on total drug, which is the acknowledged weakness of the filter (§2).

3.3 Everything else: not evaluable, and we say so

Drug	Filter 3 verdict	Why
Carfilzomib	Not evaluable	No published IC50 in an aneuploidy-stratified model; Ippolito used bortezomib and MG132. Cmax is known (2.89 μ M at 56 mg/m ²) but there is no valid denominator
Ixazomib	Not evaluable	Same gap. The β 5 enzymatic IC50 (3.4 nM) is not a cellular aneuploidy-selective value
Hydroxychloroquine	Not evaluable, and mechanistically weakened	Tang's hit was <i>chloroquine</i> , not HCQ — and chloroquine did not differentially inhibit the human CIN lines in that same paper
Everolimus / sirolimus	Not evaluable, and possibly counterproductive	See §3.4

Substituting an IC50 from an unstratified tumour line would manufacture a ratio without meaning. We leave those cells empty rather than fill them.

3.4 The antagonism nobody should walk into

A team building a combination might reach for an mTOR inhibitor alongside a proteasome inhibitor: both touch protein homeostasis, so synergy feels intuitive. The evidence points the other way.

Aneuploid and CIN cells are vulnerable to proteasome inhibition **because** they carry a translational burden. Reducing protein synthesis relieves that burden — and the ovarian carcinoma work shows exactly that mechanism operating as *resistance*. Ovarian cancer cells kept their sensitivity to proteasome inhibitors in adherent culture but “**acquired resistance as spheroids ... by downregulating protein synthesis via mTORC1 suppression**”; conversely, PTEN loss drove constitutive mTORC1 activity and **resensitized** the spheroids to proteasome inhibition, in vitro and in vivo (PMID 31530568).

Two boundaries on that evidence, since we are the ones importing it. It is ovarian carcinoma spheroids, not constitutional aneuploidy. And the paper did **not** test rapalogues — the antagonism below is our inference from its mechanism, not its result.

The objection this raises against our own candidate, stated before a reviewer does. The same paper notes that this protective mechanism “may explain the overall poor response to bortezomib in clinical trials of patients with advanced ovarian cancer.” Bortezomib has therefore *failed* in at least one solid-tumour setting, and any honest reading has to carry that. Our answer is that the proposed failure mode is context-specific — a spheroid/ascites configuration suppressing translation — and that the same paper identifies a responsive subpopulation defined by mTORC1 status. It is a reason to measure the response in patient cells before believing it, which is what §6 does; it is not a reason the mechanism is wrong.

Prediction, stated so that it can be falsified: everolimus or sirolimus would antagonise bortezomib in aneuploid cells rather than potentiate it. Any screen should test that combination explicitly and expect a negative interaction.

A tension we flag rather than hide. The same BubR1 mouse work reports that predisposition to sarcopenia correlates with **mTORC1 hyperactivity** (PMID 31738183) — and this child has skeletal muscle atrophy. So an independent, muscle-directed rationale for mTOR inhibition in MVA exists, and it points the opposite way from our antagonism prediction. The two are not strictly in conflict: one concerns proteotoxic killing of aneuploid cells, the other a progeroid muscle phenotype. But anyone testing the combination should know both exist, and should not read a muscle-directed mTOR rationale as support for pairing it with a proteasome inhibitor.

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muscle phenotype. But anyone testing the combination should know both exist, and should not read a muscle-directed mTOR rationale as support for combining it with a proteasome inhibitor.

4. Filter 4: safety in *this* child — where the candidate stops

Bortezomib clears the pharmacology. It does not clear this patient today.

Concern	Data	Relevance here
Peripheral neuropathy	25/140 (18%) of paediatric patients, motor neuropathy 11/140 (8%) (FDA BPCA Clinical Review, NDA 021602, 2015). In adults, grade ≥ 2 in 39% and grade ≥ 3 in 15% by IV route (FDA label, NDA 206927 s002, 2022 — a different source from the PK figures above)	This child has skeletal muscle atrophy (HP:0003202). Superimposing a motor neuropathy on reduced neuromuscular reserve is the decisive objection
Thrombocytopenia, neutropenia, infection	FDA label	Prior chemotherapy for ERMS; marrow reserve unknown
Gastrointestinal toxicity, hypotension	FDA label	Failure to thrive (HP:0001508) — anything suppressing intake is costly
Renal	PK not clinically altered by renal impairment per label	Nephrocalcinosis (HP:0000121) still requires renal function, hydration and electrolytes to be established first

Bortezomib is therefore not proposed as prophylaxis or as chronic therapy for this child.

But filter 4 is patient-specific, not universal — and that is the point. For a patient with MVA **and an active malignancy**, where cytotoxic therapy is on the table regardless and the comparator is conventional chemotherapy rather than nothing, the balance inverts. In that setting a proteasome inhibitor is not an exotic addition: it is an approved agent that targets the best-characterised downstream consequence of

the underlying lesion — proteotoxic dependency — supported by a human clinical signal tying response to aneuploidy burden. It does not correct the checkpoint defect, and we do not claim it does.

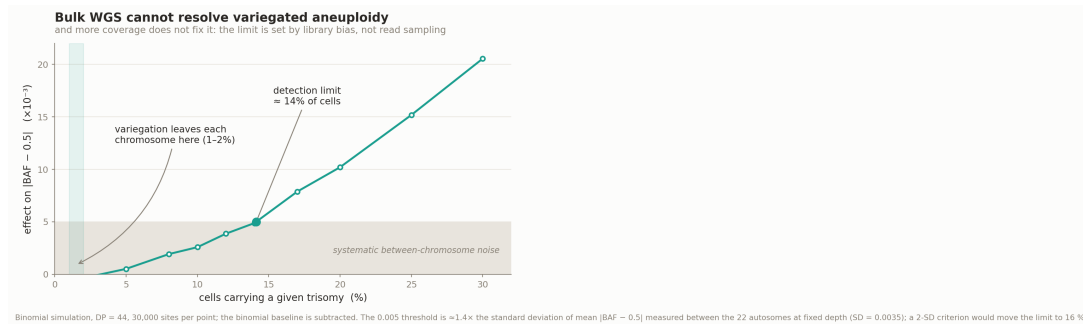
This child survived an embryonal rhabdomyosarcoma, and the risk is gene-specific: “individuals with biallelic TRIP13 or BUB1B mutations have a high risk of embryonal tumors” (PMID 28553959), and an E-RMS arising in an MVA case with biallelic BUB1B has been characterised directly (PMID 16182441). **The realistic clinical question is not “what do we give him today” but “what do we reach for if there is a second tumour” — and that question deserves an answer prepared in advance, with its evidence and its threshold, rather than improvised under pressure.**

There is a second, immediate implication. Spindle poisons — vincristine, taxanes — depend on a competent mitotic checkpoint. With reduced BubR1 that arrest is less robust; a case of rhabdomyosarcoma with PCS/MVA treated with **reduced-intensity chemotherapy** is on record (PMID 31184400 — a single case, not a guideline and not a safety estimate; PubMed carries no abstract for it, so we cite only what its title states and do not name the regimen). This is a **relative** contraindication weighed against probability of cure, never a reason to withhold curative therapy. If a mitotic-poison-sparing regimen is ever needed, there is **no validated replacement agent**: a proteasome inhibitor would be a bench hypothesis to test first, not a substitution anyone could make at the bedside.

We state the boundary explicitly: this is a hypothesis for a tumour board, not a prescription.

5. Why any endpoint must be measured per cell

Any trial of the above needs a readout for **aneuploidy burden**. We asked whether the data this hackathon provides could supply one, and computed the answer instead of asserting it.



Detection limit of bulk WGS for mosaic aneuploidy

We are not proposing a new assay. The correct endpoints exist and are standard: metaphase karyotyping with premature chromatid separation scoring is the diagnostic criterion for MVA; the micronucleus assay is an international standard (OECD TG 487); and single-cell DNA sequencing has been used to assay chromosome copy number genome-wide since Knouse et al. 2014 (PMID 25197050). We cite that work for the method, not for our conclusion — its own finding is that aneuploidy is *rarer* in liver and brain than previously reported.

What we contribute is the number.

The analysis used the **VCF only** — no BAM, hence no GC-LOESS normalisation. Four analyses were run, of which three are refinements of the same B-allele-frequency statistic and one (depth-based dosage) is orthogonal; they are not four independent tests. In a trisomy, heterozygous sites split symmetrically, so the *mean* BAF does not move and the correct statistic is $\text{mean } |\text{BAF} - 0.5|$.

The measured standard deviation of $\text{mean } |\text{BAF} - 0.5|$ across the 22 autosomes at fixed depth (DP 40–48, 857,435 sites) is **0.0035**, so the 0.005 noise floor we use — arising from GC content, mappability and segmental duplications — is a ≈ 1.4 -SD criterion. Simulating at DP 44 against it gives a **practical detection limit of a clonal whole-chromosome trisomy present in ~14% of cells**; a conventional 2-SD criterion moves it to 16% and 3-SD to 20%. Every stricter choice makes the assay look worse, so the conclusion does not rest on the threshold.

Read sampling is not what sets that limit. The simulation subtracts the binomial baseline at DP 44, so the 0.005 threshold a signal must clear is the *measured dispersion between chromosomes* — systematic library bias, not counting noise. The depth-based dosage test points the same way by a separate route: Poisson error when averaging a whole

chromosome at 44× is ~0.03%, far below the observed deviations. These are two different statistics and we do not conflate them; both imply that sequencing deeper would barely move the limit.

As an **illustrative bound, not a measurement of this child**: if 30% of cells were aneuploid — the order of magnitude of the diagnostic criterion, not a figure from this patient — spread across 22 autosomes, each chromosome would be affected in **1–2% of cells**, and gains and losses of the same chromosome cancel in bulk. That sits **7–16× below** the 14–16% limit. And **bulk averaging erases variegation by construction**, at any depth. The cytogenetic hallmark of MVA, premature chromatid separation, leaves no trace in DNA sequence at all.

Every candidate signal we did observe traced to coverage, to segmental duplications (22q11, centromeres), or partially to GC content — partially, because our measured GC–coverage correlation was $r = 0.43$ from a sparse sample, and chr21, the largest deviation, is GC-poor. That is an insufficient measurement rather than an established explanation, and we report it as such.

Consequence: no proposal whose endpoint is “reduced aneuploidy burden” can be evaluated with these data. Any screen must measure per-cell mis-segregation.

6. The eight-week go/no-go experiment

Scoped to be executable by one laboratory in one window, not as a grant application.

Question. Are BUB1B-deficient patient cells preferentially sensitive to bortezomib at clinically achievable concentrations, and does that sensitivity come without increasing mis-segregation among the survivors?

Week	Step	Readout
1–3	Patient dermal fibroblasts from skin biopsy, plus two matched controls. Patient cells are required: RPE1 with reversine-induced aneuploidy reproduces Ippolito's own system and cannot test whether the dependency transfers to BUB1B-deficient constitutional MVA — we list it as a positive control, not a substitute. Without patient cells the go/no-go question is not answerable	Growth; baseline karyotype
2–4	Allele fate: allele-specific RT-PCR ± NMD inhibitor; quantitative western blot with N- and C-terminal antibodies	N+/C- band implies truncation; loss of both implies NMD
4–6	Bortezomib dose–response, 0–100 nM, 72 h, patient versus control	EC50 and the ratio between them
4–6	Combination arm: bortezomib + everolimus	Tests the predicted antagonism
6–8	Micronucleus assay (OECD TG 487) on survivors of every condition, plus FISH for 3–5 chromosomes	Mis-segregation rate — filter 5

Timeline caveat. Eight weeks holds only if a patient fibroblast line already exists or an explant grows at the fast end of its range; establishing $15\text{--}20 \times 10^6$ fibroblasts from a fresh biopsy typically takes 4–8 weeks on its own, which would push the readouts back by roughly a month.

Advancement criteria, fixed before the data: a ≥ 3 -fold EC50 window between patient and control cells; effect at ≤ 312 nM, the clinical Cmax; and **no increase in micronuclei among survivors**. Failing any one stops the programme.

Falsifying results. If patient fibroblasts are no more sensitive than controls, the proteotoxic dependency does not transfer from cancer aneuploidy to constitutional MVA. If bortezomib raises micronucleus frequency among survivors, it fails filter 5 regardless of how well it kills.

This design can fail, which is the point of proposing it.

7. Actionable now, without any drug

The intervention that most changes this child's prognosis today is not a molecule.

Published UK recommendations for children at high Wilms risk propose renal ultrasound every 3–4 months until age 5 (PMID 16857697); these are Wilms-risk recommendations applied by analogy, not BUB1B-specific evidence. Contemporary guidance for genomic instability syndromes emphasises individualised planning, minimising ionising radiation and management in expert centres (DOI 10.1158/1078-0432.CCR-24-1101).

Our pragmatic proposal, with the inference flagged. **Surveillance windows must be anchored to the child's current age, which is not in the dataset and must be confirmed before use:**

Period	Surveillance	Frequency
To age 7	Full abdominal ultrasound: both kidneys, liver, retroperitoneum, pelvis	Every 3 months. Extending from 5 to 7 years is our cautious inference , not validated for BUB1B
To age 10	Full paediatric and oncological examination: abdomen, nodes, head and neck, orbit, skin, genitalia, soft tissues	3-monthly to age 7, then 6-monthly. Catches superficial ERMS that renal ultrasound cannot
Post-ERMS	Relapse follow-up per tumour protocol	Coordinate to avoid duplicate imaging
Through childhood	Full blood count with differential	6-monthly is reasonable given reported leukaemia, but no evidence of benefit exists
Lifelong	Family education; annual predisposition clinic	Urgent review for mass, haematuria, abdominal distension, persistent pain, neurological deficit, bleeding or weight loss

If the child is already past 7, missed scans should not be “caught up”: a baseline assessment and an individualised decision by the predisposition team is the right course.

We do **not** propose serial AFP, repeat CT, PET-CT or routine whole-body MRI while asymptomatic — no demonstrated benefit in MVA, weighed against sedation, false positives, cost and radiation.

8. What we do not propose

- **MPS1/TTK inhibitors and reversine** — not approved, and they weaken a checkpoint that is already insufficient. Reversine is a research tool for *inducing* aneuploidy, the opposite of the goal.
- **Apcin, proTAME** — APC/C tool compounds, not marketed.
- **DCZ0415** — experimental TRIP13 inhibitor: wrong gene, not approved.
- **AICAR and 17-AAG** — genuine hits in the Tang screen, neither marketed. Useful as in vitro positive controls only.
- **Metformin** — fails filter 3 by roughly 1000-fold, as set out in §3.1.

- **Senolytics** — the BubR1 senescence evidence rests on *genetic* clearance of p16^{Ink4a} cells via the INK-ATTAC transgene (PMID 22048312), which is not a transferable drug; no agent is approved with a senolytic indication; and senescence is an anti-tumour barrier worth preserving in a cancer-predisposition syndrome.
 - **mTOR inhibitors combined with proteasome inhibition** — predicted antagonism (PMID 31530568).
 - **Bortezomib for this child today** — fails filter 4, as set out in §4.
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9. Scalability

- The **five-filter framework**, particularly filters 3 and 5, applies to any repurposing exercise in any rare disease. It is the transferable product of this work.
 - The **proteotoxic vulnerability**, *if* §6 shows it transfers, would extend most plausibly to **TRIP13**-related MVA, which shares both the tumour risk and the severity of checkpoint failure: “individuals with biallelic TRIP13 or BUB1B mutations have a high risk of embryonal tumors ... their cells display severe SAC impairment” (PMID 28553959). **It should not be assumed for CEP57**. The same source reports that MVA from biallelic CEP57 mutations “is not associated with embryonal tumors” and that those cells “show minimal SAC deficiency” — less mis-segregation should mean less proteotoxic burden, so the rationale is weaker there. No proteasome data exist for either, and §10 states that transfer to MVA1 itself is unproven.
 - The **gene panel** (BUB1B, CEP57, TRIP13, CENATAC, MAD1L1, MAD2L1BP, CEP192, BUB1, SMC5, TRIM37, CENPE) and the artefact controls used in the mosaicism analysis transfer to any MVA or PCS workup.
 - The **micronucleus → scDNA-seq endpoint** applies to mitotic CIN disorders. It does **not** transfer to Fanconi anaemia or Bloom syndrome, whose diagnostic assays are DEB/MMC chromosome breakage and sister-chromatid exchange respectively — a different lesion, not substitutable.
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10. Limitations

- **Phase is not established.** No parental samples, and the variants lie ~11 kb apart, beyond read-backed phasing for short reads. Two heterozygous variants in a singleton are a priori 50/50 cis or trans; rarity does not phase them. What supports *trans* is that the child is affected, that MVA1 is recessive, that truncating-plus-missense is the recurrent pattern, and that the phenotype matches. It remains a diagnosis pending segregation.
 - **p.Asn1002Lys has no functional assay**, and its apparent ClinVar entry is for a different nucleotide.
 - **The bortezomib EC50 is a conservative upper bound** read from a figure panel rather than a tabulated value, and the Cmax is a label interval rather than a point estimate; the ratio of 5.8–7.8 should be read as an order-of-magnitude statement computed on total, not free, drug.
 - **Ippolito et al. studied cancer aneuploidy**, not constitutional variegated mosaicism. Whether the proteotoxic dependency transfers is precisely what the proposed experiment tests; we have not assumed it.
 - **No trial specific to MVA1** was found in ClinicalTrials.gov, Open Targets, ChEMBL or CMap/LINCS. Absence of an interface result is not proof of absence.
 - **The child's current age is unknown to us**, and the surveillance schedule depends on it.
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A closing note

The family of this child published their son's genome so that strangers might help. They will read what comes back.

We have tried to write something useful rather than something reassuring. The honest answer to “what drug should he take today” is none — and we show why the obvious candidate cannot work rather than proposing it softly. The useful answer is a mechanism that matches his disease, an approved drug that reaches the required

concentration, an explicit statement of the condition under which it would become appropriate, and an experiment that could rule it out in eight weeks.

That is not a cure. It is a prepared answer to a question this family may unfortunately have to ask.

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